

UNIVERSITY OF GEORGIA

Clustering Algorithms Finding Structure Without Labels

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Outline

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Introduction

What clustering is

2

Principles

k-means, hierarchical, DBSCAN, GMM by hand

3

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Which algorithm suits which data

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Case Study

Athens intersections by crash mix

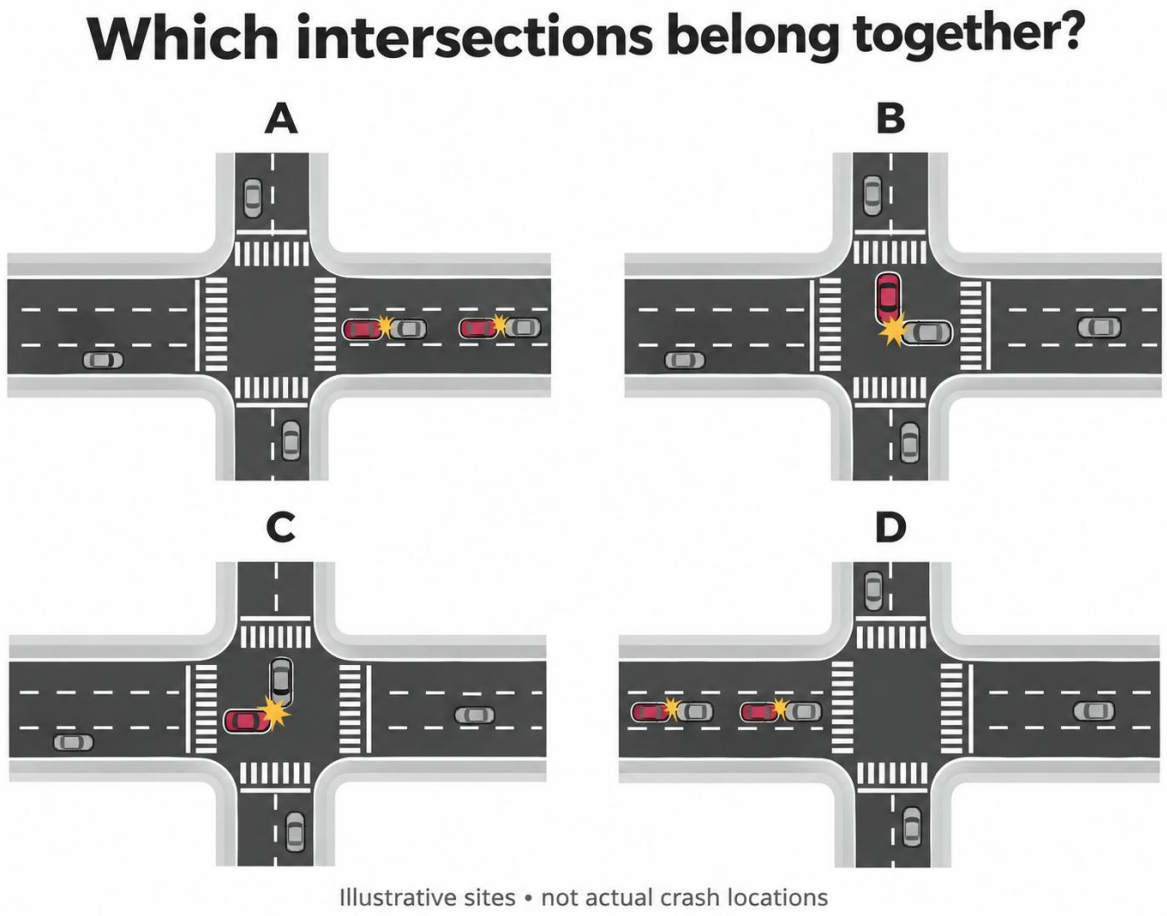
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Summary

Choosing a method

Notebook: Clustering_Lecture.ipynb: the methods and their examples (parts 1–3).

A Question Without Labels



THE DATA

189 intersections, 2021–2025

6,755 crashes

11 crash-mix features per intersection

THE QUESTION

Which intersections have a similar crash pattern, so they can share a treatment plan?

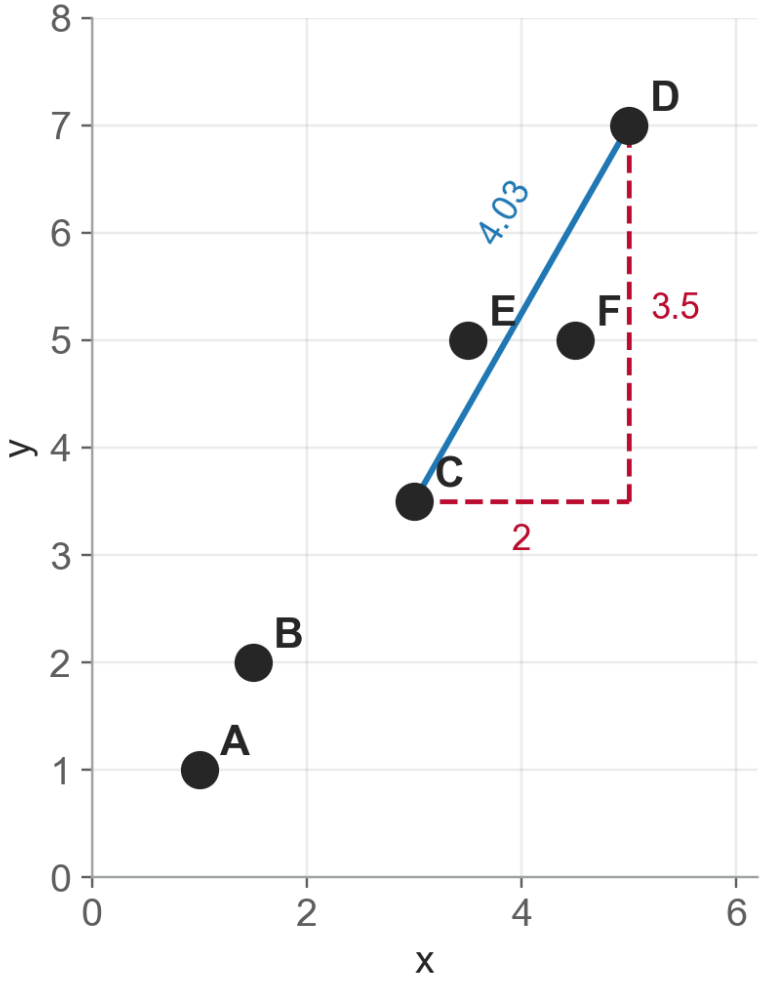
No labels: nobody has told us the right groups.

Clustering: Three Questions Behind Every Method

1 Similarity	2 A cluster	3 How many
How is it measured? distance: Euclidean, Manhattan	What does one look like? centre, tree, dense region, distribution	Is k given or found?

Family	Idea of a cluster	Example algorithms	k given?
Centroid-based	points close to a centre	k-means	yes
Hierarchical	a tree of nested groups	agglomerative (single, complete, Ward)	cut the tree
Density-based	dense regions, sparse gaps	DBSCAN, OPTICS	no
Distribution-based	drawn from one distribution	Gaussian mixture (EM)	yes
Graph-based	well-connected part of a graph	spectral clustering	yes

Distance: What "Similar" Means



Point	A	B	C	D	E	F
x	1	1.5	3	5	3.5	4.5
y	1	2	3.5	7	5	5

EUCLIDEAN

$$d(C, D) = \sqrt{2^2 + 3.5^2} = 4.03$$

MANHATTAN

$$d_1(C, D) = |2| + |3.5| = 5.5$$

Units: with real data, scale every feature first.

k-means: Objective and Algorithm

Objective

$$\mathrm{SSE} = \sum_{j=1}^{k} \sum_{x_i \in C_j} \left\| x_i - \mu_j \right\|^2$$

μ_j: mean of the points in cluster C_j

1 Initialise → **2 Assign** → **3 Update** → **4 Stop?**

1 Initialise	2 Assign	3 Update	4 Stop?
choose k centres	each point → its nearest centre	each centre → mean of its points	no point changed cluster

changed → repeat

Note: the SSE never increases, but the final result can depend on the starting centres (local minimum).

k-means by Hand: Iteration 1

ASSIGN: SQUARED DISTANCES

Point	d² to μ1 = (1, 1)	d² to μ2 = (5, 7)	Cluster
A (1, 1)	0	52	1
B (1.5, 2)	0.25 + 1 = 1.25	12.25 + 25 = 37.25	1
C (3, 3.5)	4 + 6.25 = 10.25	4 + 12.25 = 16.25	1
D (5, 7)	52	0	2
E (3.5, 5)	6.25 + 16 = 22.25	2.25 + 4 = 6.25	2
F (4.5, 5)	12.25 + 16 = 28.25	0.25 + 4 = 4.25	2

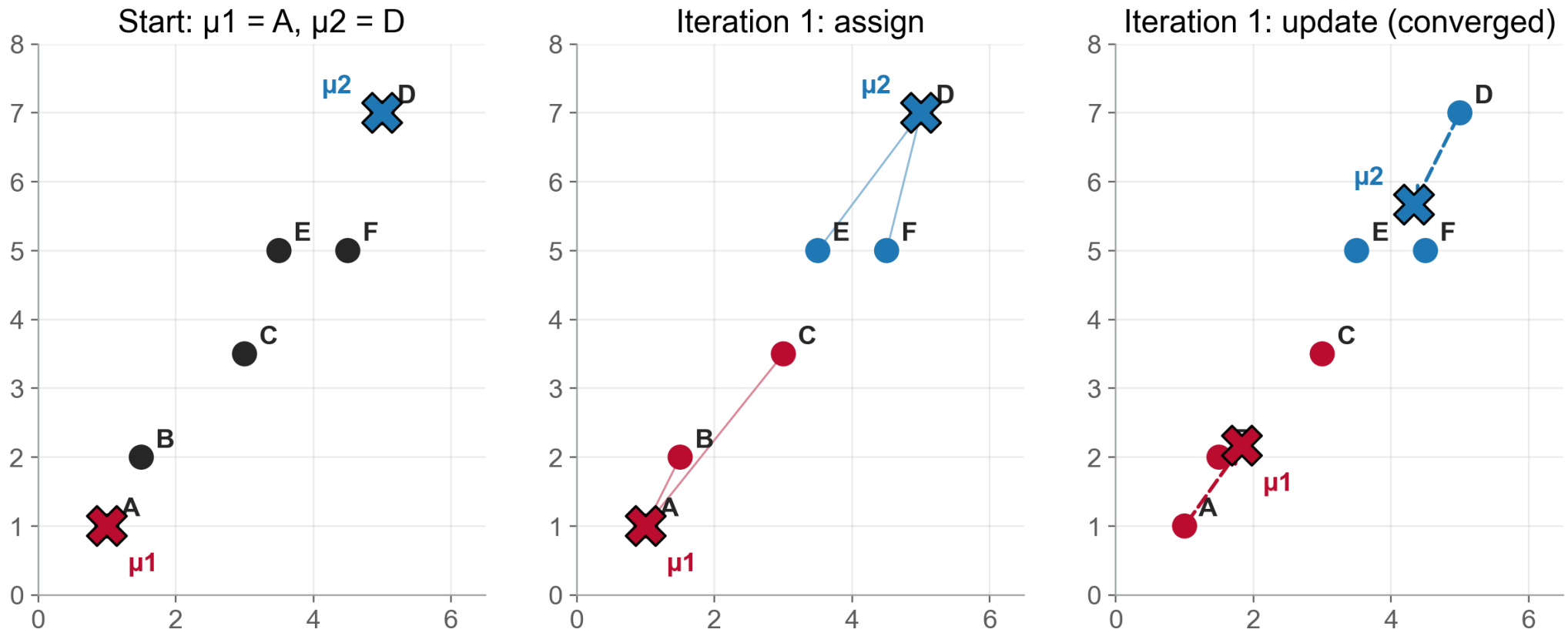
UPDATE: NEW MEANS

$$\boldsymbol{\mu}_1 = \frac{1}{3}(A + B + C) = (1.83, 2.17)$$

$$\boldsymbol{\mu}_2 = \frac{1}{3}(D + E + F) = (4.33, 5.67)$$

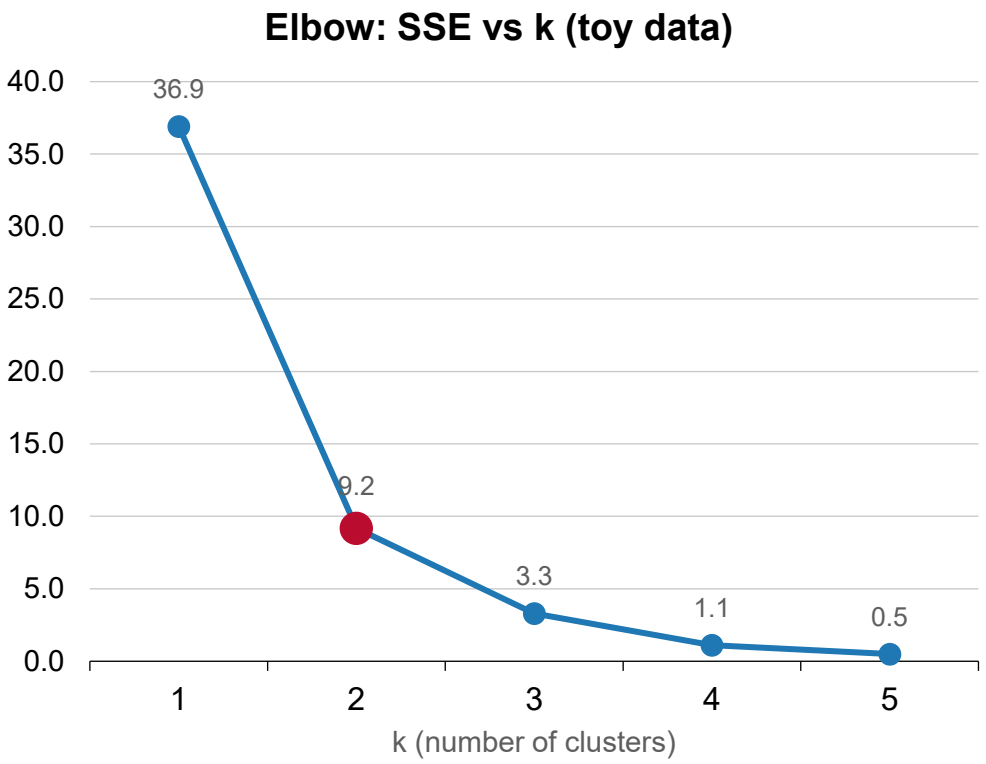
Question: with the new centres, does C change cluster?

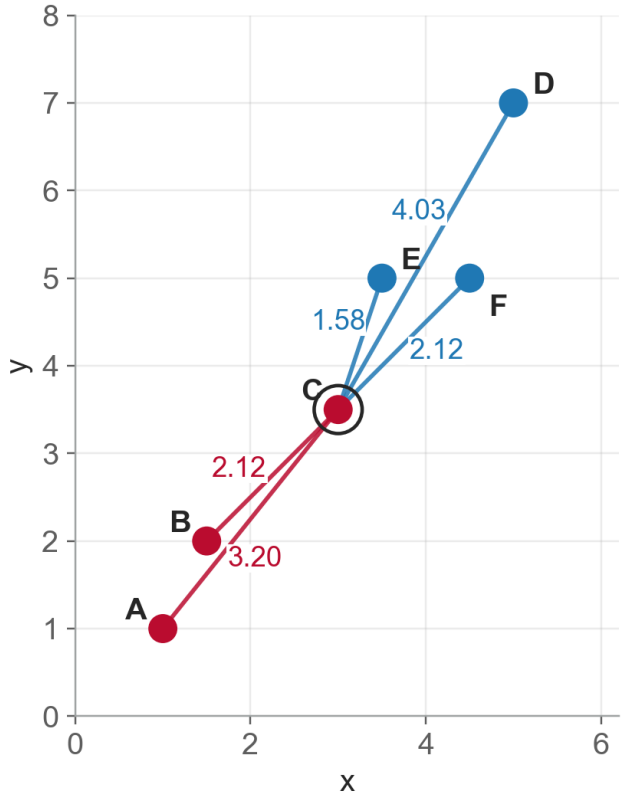
k-means by Hand: Convergence



Iteration 2: C: $d^2 = 3.14$ to μ1 vs 6.47 to μ2. No point moves → converged. SSE = 9.17

Choosing k and Checking Each Point





$$s = \frac{b - a}{\max(a, b)}$$

$$a = \frac{3.20 + 2.12}{2} = 2.66$$

$$b = \frac{4.03 + 1.58 + 2.12}{3} = 2.58$$

s(C) = −0.03

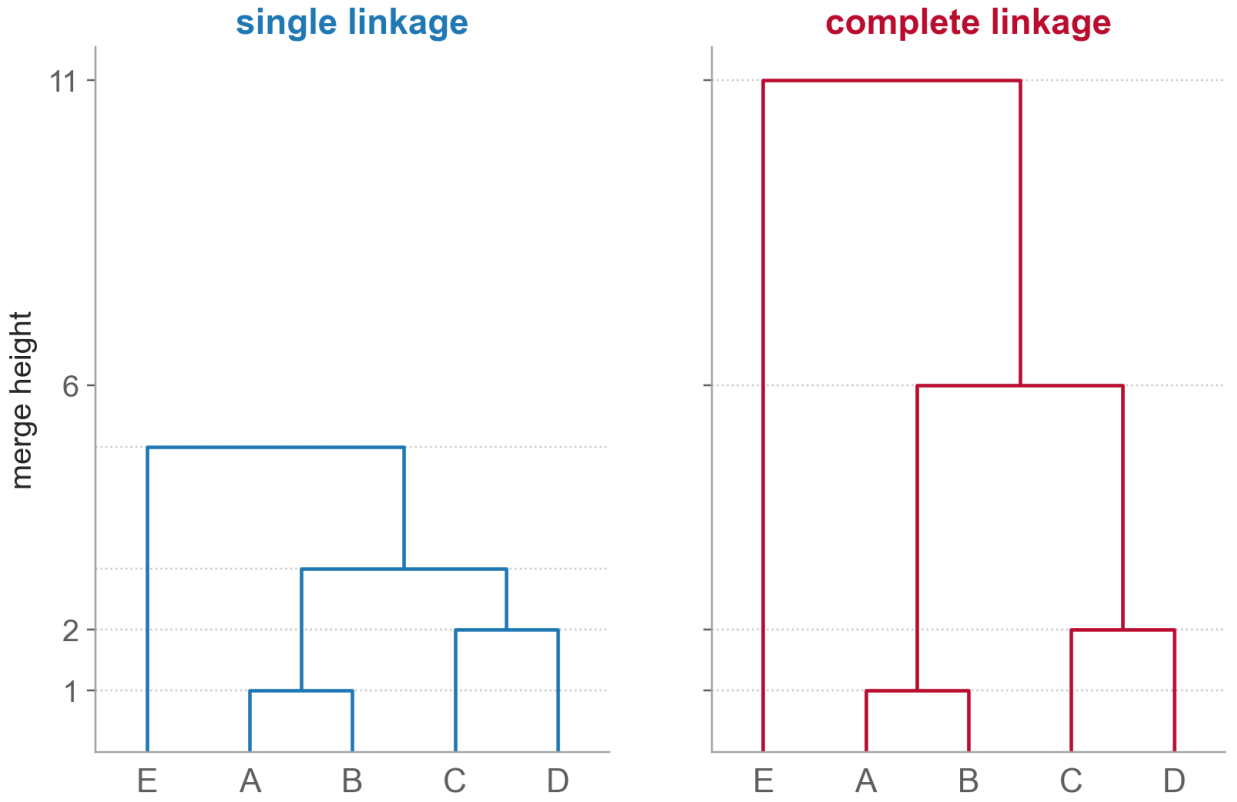
Silhouette: close to 1: well inside its cluster; near 0: on a border; below 0: closer to another cluster. Mean here = 0.49

Hierarchical Clustering: Build a Tree

Linkage	Distance between groups	Tends to give
single	closest pair	long chains
complete	farthest pair	compact groups
average	mean of all pairs	in between
Ward	increase in SSE	round groups

Points: A=1 B=2 C=5 D=7 E=12

Step 3, single: {A,B}–{C,D} = |2 − 5| = 3
Step 3, complete: {A,B}–{C,D} = |1 − 7| = 6

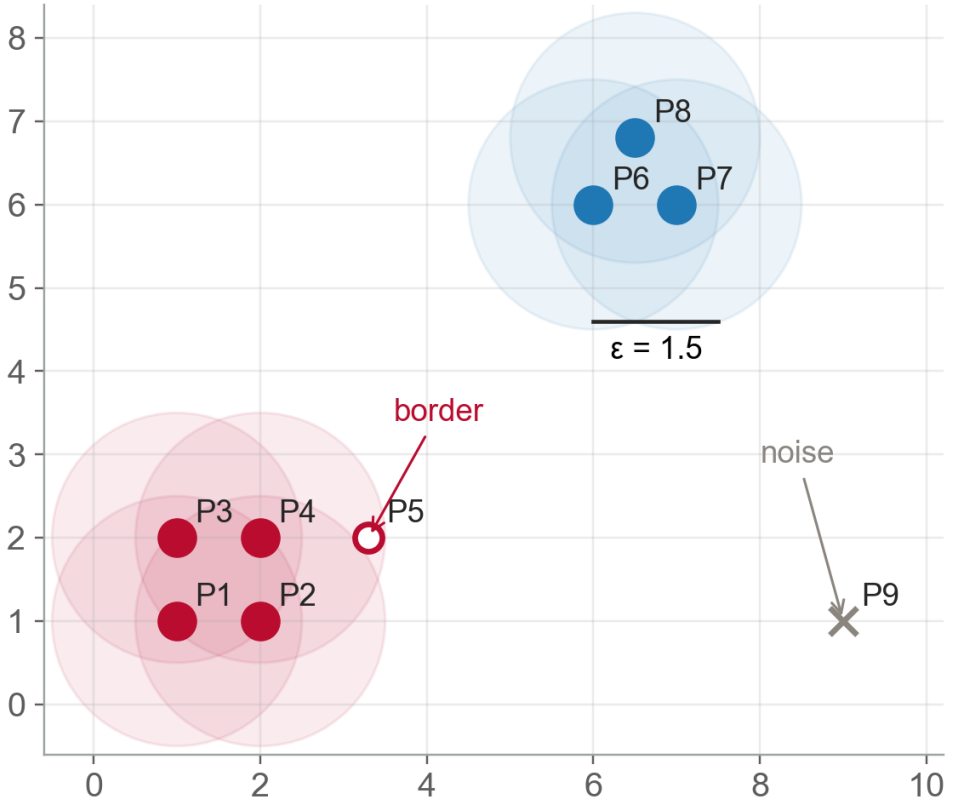


Cut the tree: a horizontal cut gives the clusters; its height sets how many.

DBSCAN: Clusters as Dense Regions

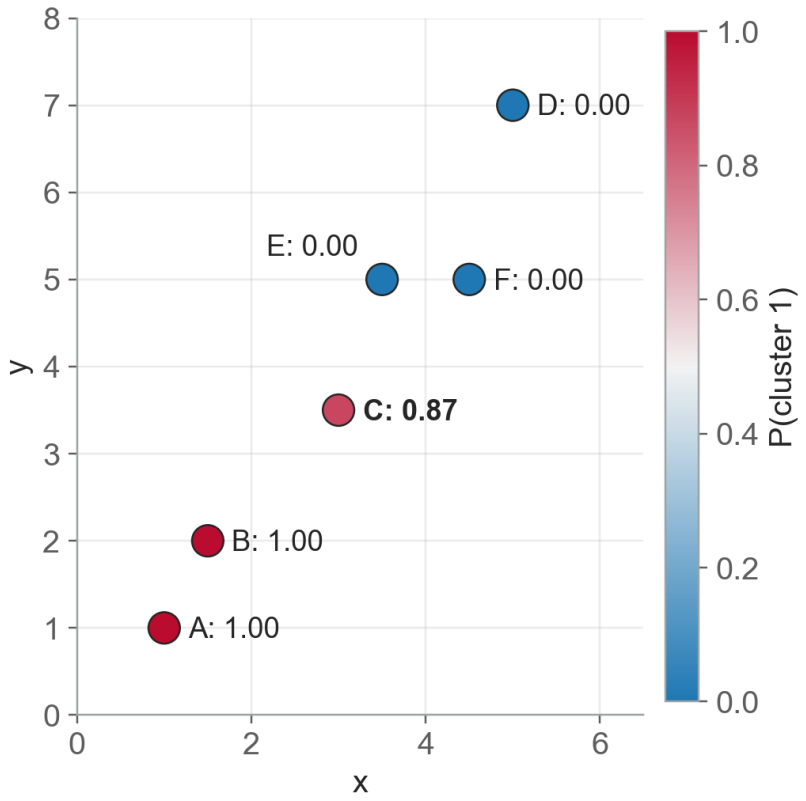
Core	at least minPts points within ε (itself included)
Border	not core, but within ε of a core point
Noise	neither core nor border

ε = 1.5, minPts = 3: clusters {P1…P5} and {P6, P7, P8}; P9 is noise.



k not given: the number of clusters follows from ε and minPts, and points may be left as noise.

Gaussian Mixtures: Soft Clustering

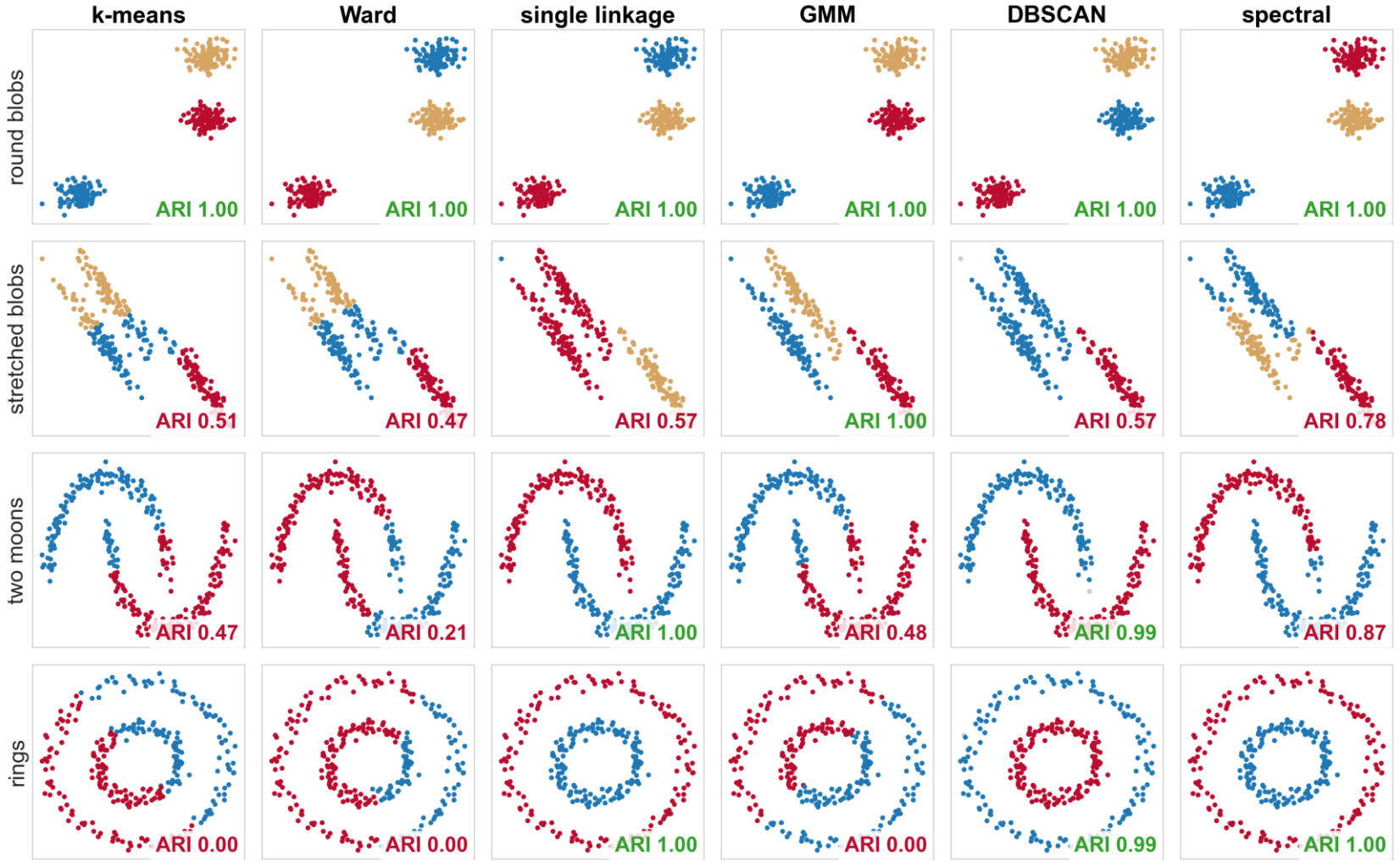


$$p(\boldsymbol{x}) = \sum_{j=1}^{k} \pi_j \mathcal{N}\left(\boldsymbol{x} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j\right)$$

	k-means	GMM (EM)
Assign	nearest centre (0/1)	E-step: probabilities
Update	means	M-step: weights, means, covariances
Cluster shape	round	elliptical

Toy data: P(C in cluster 1) = 0.87; all other points ≥ 0.99.

Which Algorithm Suits Which Data?



OBSERVATIONS

k-means, Ward: round groups of similar size

GMM: elliptical groups

Single, DBSCAN, spectral: connected shapes

ARI: 1 = same as the true groups, about 0 = chance level

Case Study: Intersection Types

Feature	Share of the intersection's crashes that…
rear end, angle, sideswipe	had this manner of collision
left turn	involved a left-turning vehicle
night, wet road	happened 19:00–06:59 / on a wet surface
signal, stop sign	had this control reported on the crash report
failed to yield	had it as the first contributing factor
injury, ped./cyclist	involved an injury / a pedestrian or cyclist

DATA

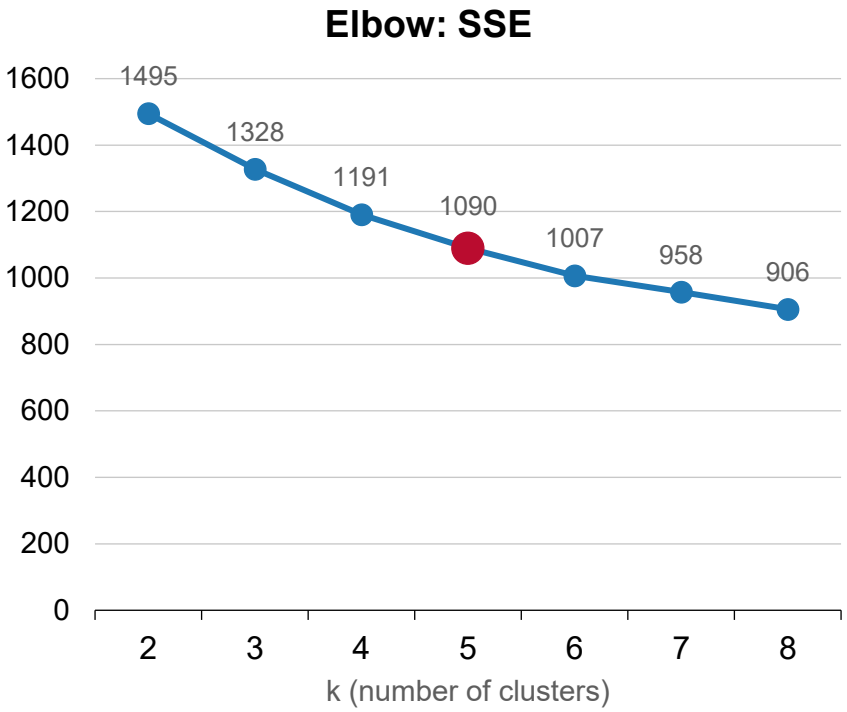
189 intersections with ≥ 15 crashes

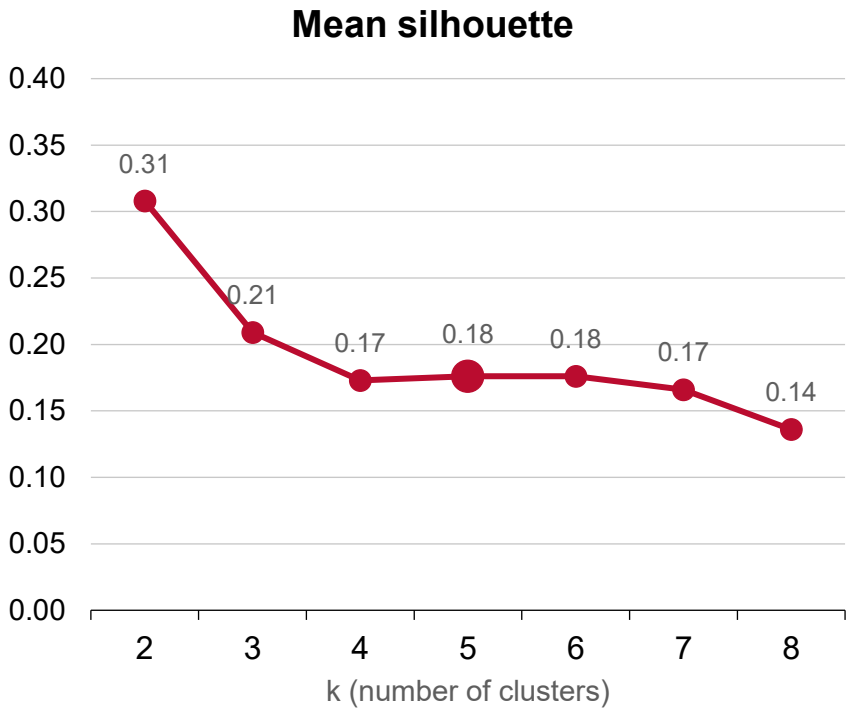
Athens-Clarke County, 2021–2025

11 features, each a share (0–1)

Features → **Scale** → **Choose k** → **Cluster** → **Interpret** → **Check**

Scale, Then Choose k





READING

No sharp elbow

Silhouette ≤ 0.31

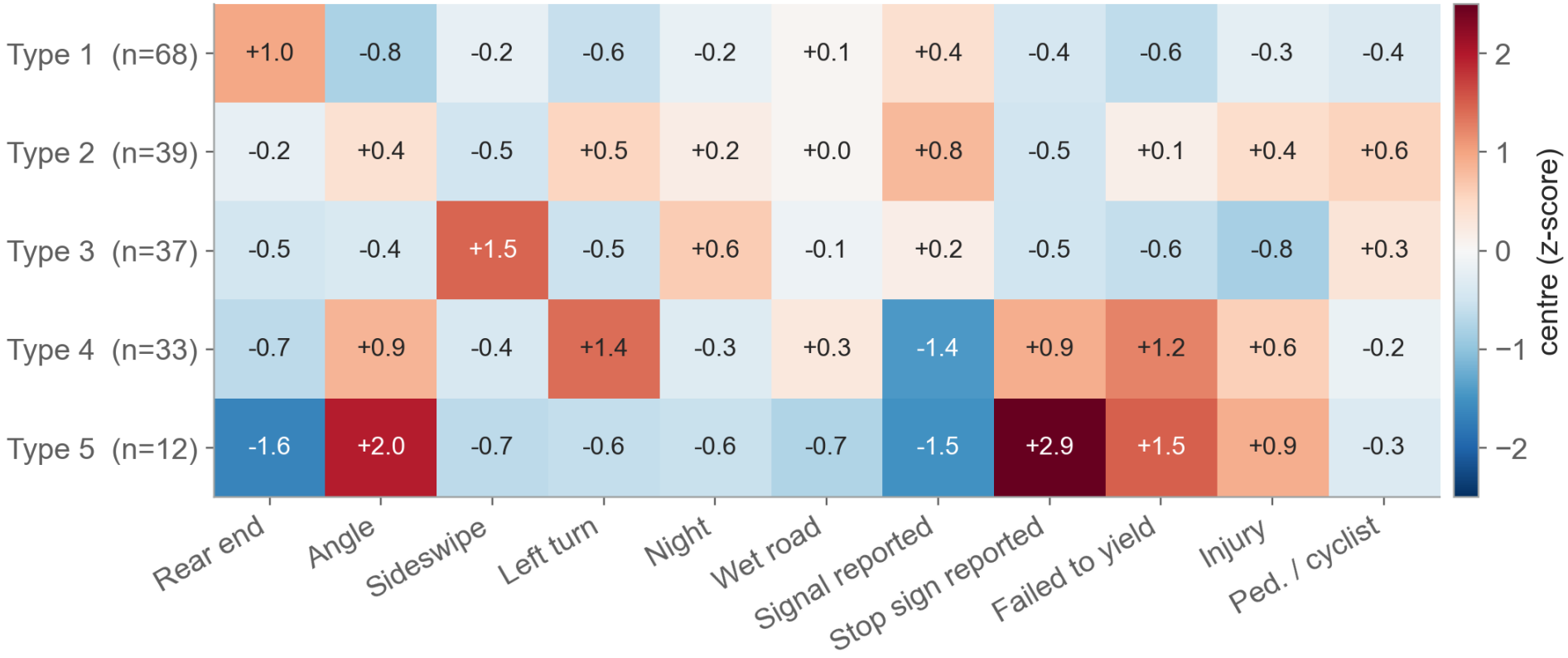
Gradual change between sites

k = 5

Scaling (z-score) $z = \frac{x - \bar{x}}{s}$

$\bar{x}$, s: mean and standard deviation of the feature

Five Intersection Types



Reading: each row is a cluster centre (z-score); red = above the average intersection, blue = below.

From Types to Countermeasure Families

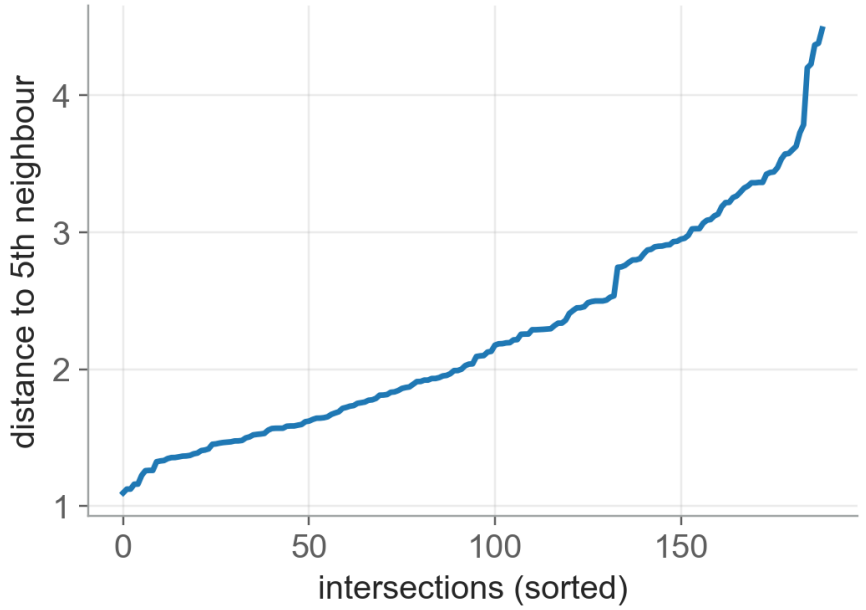
Type	Sites	Stands out (type vs all intersections)	Countermeasure family
Type 1	68	Rear end 60% vs 41%; Signal reported 59% vs 47%	Signal timing, advance warning
Type 2	39	Signal reported 71% vs 47%; Ped. / cyclist 3% vs 1%	Crosswalks, refuge islands
Type 3	37	Sideswipe 30% vs 14%; Night 28% vs 22%	Lane assignment, markings
Type 4	33	Left turn 42% vs 21%; Failed to yield 39% vs 20%	Protected left-turn phasing
Type 5	12	Stop sign reported 69% vs 10%; Angle 78% vs 40%	Stop-control package

Do Other Methods Agree?

AGREEMENT BETWEEN METHODS (ARI)

	k-means	Ward	GMM
k-means	1.00	0.53	0.39
Ward	0.53	1.00	0.35
GMM	0.39	0.35	1.00

Moderate agreement: distinct types are stable; boundaries between the others are soft.



DBSCAN (minPts 5): ε 2.0 → 2 clusters, 67 noise · ε 2.5 → 1 cluster, 35 noise · ε 3.0 → 1 cluster, 13 noise

Summary

Method	Idea	k given?	Suits	Sensitive to
k-means	nearest centre, minimise SSE	yes	round groups, large data	start, scaling, outliers
Hierarchical	merge closest groups	cut tree	seeing the structure	linkage, $O(n^2)$ memory
DBSCAN	dense regions + noise	no	any shape, noise	ε, varying density
GMM	mixture of Gaussians	yes	elliptical groups	local optima

Features	decide what can be found
Scale	before measuring distance
Check	silhouette, other methods

Thank You

Questions and comments are welcome

Notebook: Clustering_Lecture.ipynb · based on the SensorAI tutorials (github.com/wsonguga/SensorAI)



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